

LTMLE Project Outputs

Laura Aleksonis, Siyaa Bhargava, Xhemka Elezi

2026-04-18

```
#install.packages('ggplot2')
library(ggplot2)
cat('Packages retrieved!')

## Packages retrieved!
```

Retrieval of Data and Results

The working directory should be set to a folder containing all relevant items, including a folder named “Datasets” where the dataset of interest resides.

```
## Data retrieved!
```

Dataset Statistics

The threshold value used to categorize yards gained was the median of Yards gained per season in both t1 and t2.

```
## Summary of Yards Gained in t1:

##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   -11.0   65.0   197.0   266.2   368.0   2094.0

##
## Summary of Yards Gained in t2:

##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   -9.00   93.75   226.00   319.50   440.00   1948.00

##
## -      -      -

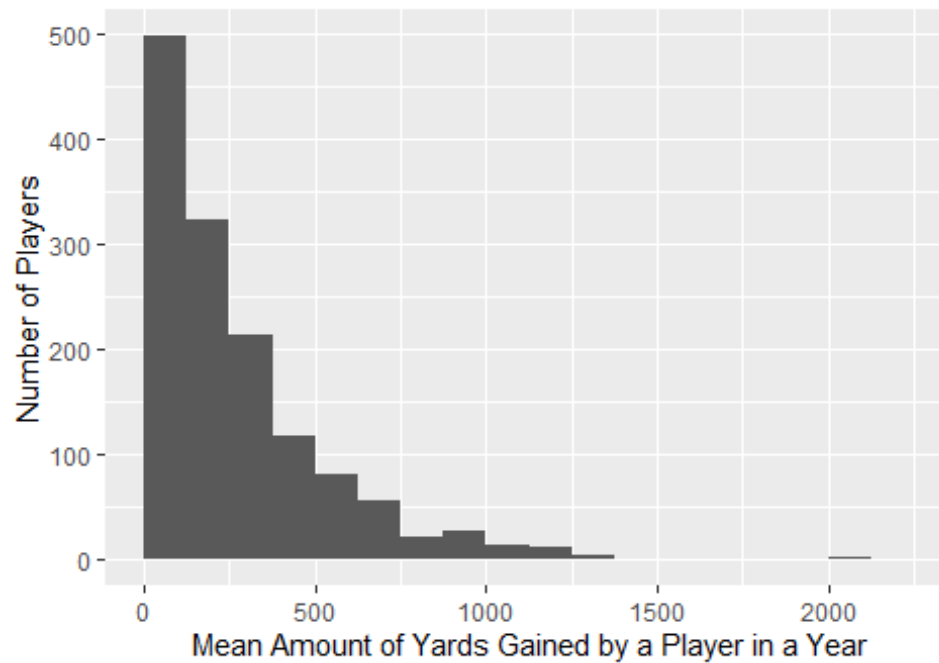
##
## Summary of Yard Categorizations:

##
##           t2
## t1           At or above median    Below median
##   At or above median           263           251
##   Below median                 169           689

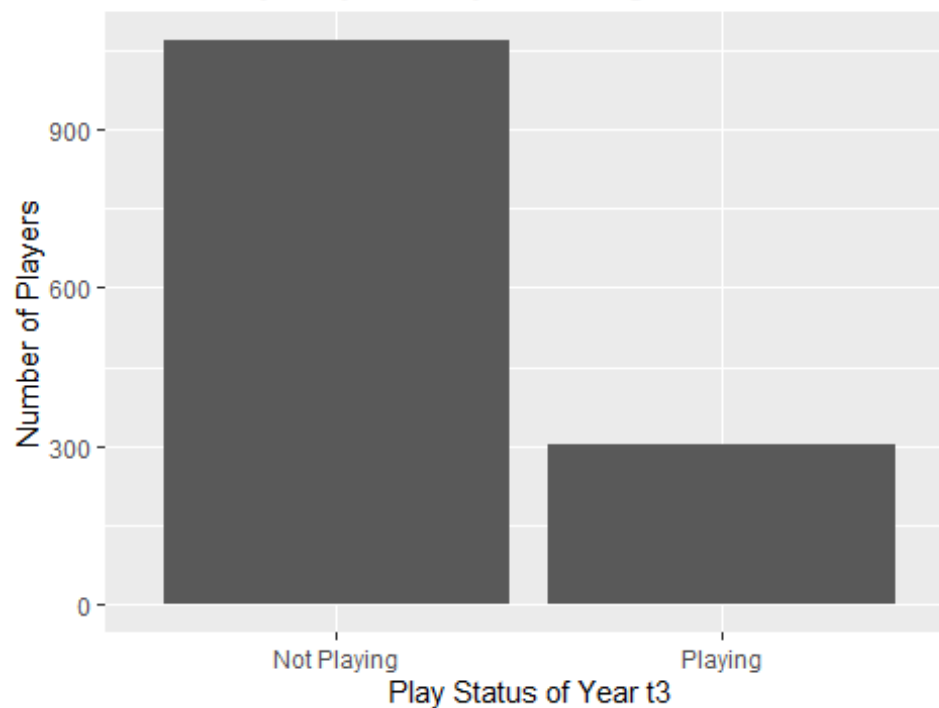
## Sample size: 1372 football players
```

Histogram of Yards Gained in a Year

Values determined for each player as the average of yards gained



How Many Players Played During their t3?



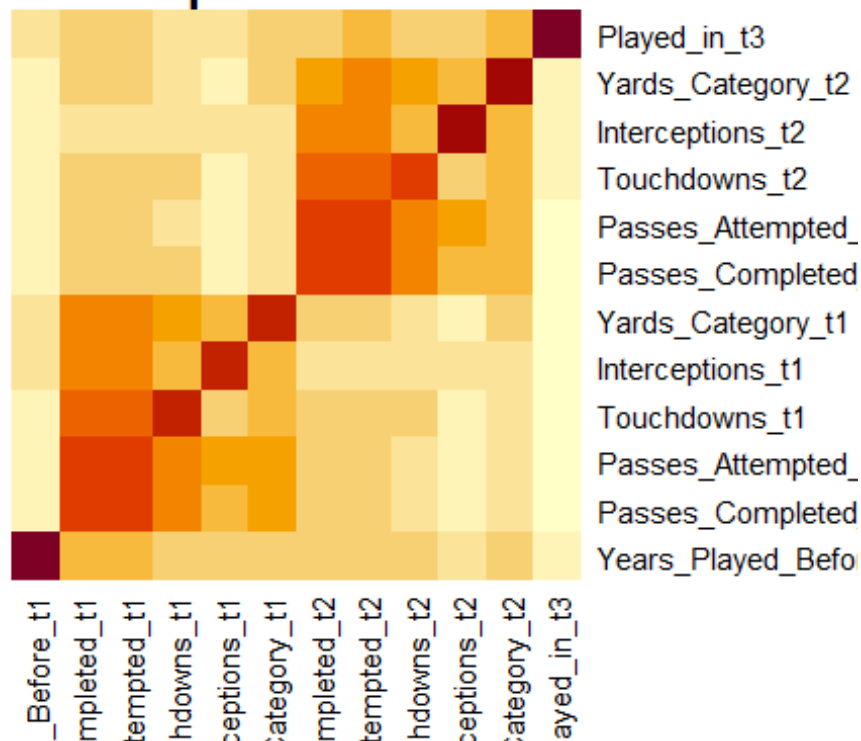
```
## Correlations between all variables included in maximal model:
```

```
##           Years_Played_Before_t1 Passes_Completed_t1
## Years_Played_Before_t1           1.00000000          0.22588328
```

## Passes_Completed_t1	0.22588328	1.00000000	
## Passes_Attempted_t1	0.21673401	0.98423703	
## Touchdowns_t1	0.17946614	0.79740102	
## Interceptions_t1	0.13494367	0.60749495	
## Yards_Category_t1	0.19595507	0.61875918	
## Passes_Completed_t2	0.19659314	0.43357376	
## Passes_Attempted_t2	0.18259565	0.42347298	
## Touchdowns_t2	0.15566246	0.37027957	
## Interceptions_t2	0.07372086	0.23173247	
## Yards_Category_t2	0.12254547	0.29801333	
## Played_in_t3	-0.08722860	0.01080638	
##	Passes_Attempted_t1	Touchdowns_t1	Interceptions_t1
## Years_Played_Before_t1	0.21673401	1.794661e-01	0.134943667
## Passes_Completed_t1	0.98423703	7.974010e-01	0.607494954
## Passes_Attempted_t1	1.00000000	7.710420e-01	0.645177893
## Touchdowns_t1	0.77104199	1.000000e+00	0.413907185
## Interceptions_t1	0.64517789	4.139072e-01	1.000000000
## Yards_Category_t1	0.62460371	5.203051e-01	0.428241903
## Passes_Completed_t2	0.41040532	3.861187e-01	0.208367495
## Passes_Attempted_t2	0.40639518	3.772598e-01	0.208638580
## Touchdowns_t2	0.34525074	3.495640e-01	0.157476443
## Interceptions_t2	0.23159389	1.942443e-01	0.149093789
## Yards_Category_t2	0.29042063	2.697283e-01	0.158269586
## Played_in_t3	0.01340713	-7.260844e-05	-0.008892665
##	Yards_Category_t1	Passes_Completed_t2	
## Years_Played_Before_t1	0.195955070	0.1965931	
## Passes_Completed_t1	0.618759179	0.4335738	
## Passes_Attempted_t1	0.624603707	0.4104053	
## Touchdowns_t1	0.520305118	0.3861187	
## Interceptions_t1	0.428241903	0.2083675	
## Yards_Category_t1	1.000000000	0.3206193	
## Passes_Completed_t2	0.320619256	1.0000000	
## Passes_Attempted_t2	0.317526190	0.9851140	
## Touchdowns_t2	0.255925383	0.8359285	
## Interceptions_t2	0.184139756	0.5945439	
## Yards_Category_t2	0.327960027	0.5704917	
## Played_in_t3	0.005498069	0.1138246	
##	Passes_Attempted_t2	Touchdowns_t2	Interceptions_t2
## Years_Played_Before_t1	0.1825957	0.1556625	0.07372086
## Passes_Completed_t1	0.4234730	0.3702796	0.23173247
## Passes_Attempted_t1	0.4063952	0.3452507	0.23159389
## Touchdowns_t1	0.3772598	0.3495640	0.19424428
## Interceptions_t1	0.2086386	0.1574764	0.14909379
## Yards_Category_t1	0.3175262	0.2559254	0.18413976
## Passes_Completed_t2	0.9851140	0.8359285	0.59454389
## Passes_Attempted_t2	1.0000000	0.8079923	0.64050748
## Touchdowns_t2	0.8079923	1.0000000	0.42448733
## Interceptions_t2	0.6405075	0.4244873	1.00000000
## Yards_Category_t2	0.5745603	0.4879373	0.39109478
## Played_in_t3	0.1177342	0.1135409	0.05927440

```
##                                Yards_Category_t2  Played_in_t3
## Years_Played_Before_t1          0.1225455 -8.722860e-02
## Passes_Completed_t1             0.2980133  1.080638e-02
## Passes_Attempted_t1             0.2904206  1.340713e-02
## Touchdowns_t1                   0.2697283 -7.260844e-05
## Interceptions_t1                 0.1582696 -8.892665e-03
## Yards_Category_t1                0.3279600  5.498069e-03
## Passes_Completed_t2             0.5704917  1.138246e-01
## Passes_Attempted_t2             0.5745603  1.177342e-01
## Touchdowns_t2                   0.4879373  1.135409e-01
## Interceptions_t2                 0.3910948  5.927440e-02
## Yards_Category_t2                1.0000000  1.188065e-01
## Played_in_t3                    0.1188065  1.000000e+00
```

Heatmap of Correlations



Comparison Statistics

Summary Statistics

```
## OUTPUT: Maximal Model Statistics
```

```
##                                Statistics
## Value_of_abar      ATE_Estimate ATE_P_Value RR_Estimate RR_P_Value
## list(c(1, 1), c(0, 0))  0.06808748  0.5230688  1.407614  0.5622164
## list(c(1, 0), c(0, 0))  0.09103050  0.5294350  1.544965  0.5308068
## list(c(0, 1), c(0, 0))  0.18384651  0.1456500  2.100620  0.2102432
##                                Statistics
```

```
## Value_of_abar          OR_Estimate OR_P_Value
## list(c(1, 1), c(0, 0))    1.532917 0.5527577
## list(c(1, 0), c(0, 0))    1.734523 0.5294833
## list(c(0, 1), c(0, 0))    2.695571 0.1874924

##
## - - -
##
## OUTPUT: Minimal Model Statistics

##
##                               Statistics
## Value_of_abar          ATE_Estimate ATE_P_Value RR_Estimate RR_P_Value
## list(c(1, 1), c(0, 0))    0.15182652 0.074391218    2.558279 0.22081663
## list(c(1, 0), c(0, 0))    0.04462753 0.679897552    1.458037 0.68681823
## list(c(0, 1), c(0, 0))    0.34968331 0.003625251    4.588992 0.04960263
##
##                               Statistics
## Value_of_abar          OR_Estimate OR_P_Value
## list(c(1, 1), c(0, 0))    3.075654 0.19144624
## list(c(1, 0), c(0, 0))    1.533880 0.68543693
## list(c(0, 1), c(0, 0))    7.491395 0.02731934
```

Complete Statistics: Maximal Model

```
##
## OUTPUT: Maximal Model
## (abar = list(c(1, 1), c(0, 0)) )
##
## Estimator:  tmle
## Call:
## ltmle(data = data, Anodes = c(paste0(treats, "_t1"), paste0(treats,
##   "_t2")), Lnodes = c("Passes_Completed_t1", "Passes_Attempted_t1",
##   "Touchdowns_t1", "Interceptions_t1", "Passes_Completed_t2",
##   "Passes_Attempted_t2", "Touchdowns_t2", "Interceptions_t2"),
##   Ynodes = "Played_in_t3", abar = abar)
##
## Treatment Estimate:
##   Parameter Estimate:  0.23513
##   Estimated Std Err:  0.057832
##   p-value:  4.7899e-05
##   95% Conf Interval: (0.12178, 0.34848)
##
## Control Estimate:
##   Parameter Estimate:  0.16704
##   Estimated Std Err:  0.089568
##   p-value:  0.06219
##   95% Conf Interval: (0, 0.34259)
##
## Additive Treatment Effect:
##   Parameter Estimate:  0.068087
##   Estimated Std Err:  0.10662
##   p-value:  0.52307
```

```

##      95% Conf Interval: (-0.14088, 0.27705)
##
## Relative Risk:
##   Parameter Estimate: 1.4076
##   Est Std Err log(RR): 0.58993
##                   p-value: 0.56222
##   95% Conf Interval: (0.44293, 4.4734)
##
## Odds Ratio:
##   Parameter Estimate: 1.5329
##   Est Std Err log(OR): 0.71959
##                   p-value: 0.55276
##   95% Conf Interval: (0.37411, 6.2811)
##
## -      -      -
##
## OUTPUT: Maximal Model
## (abar = list(c(1, 0), c(0, 0)) )
##
## Estimator:  tmle
## Call:
## ltmle(data = data, Anodes = c(paste0(treats, "_t1"), paste0(treats,
##   "_t2")), Lnodes = c("Passes_Completed_t1", "Passes_Attempted_t1",
##   "Touchdowns_t1", "Interceptions_t1", "Passes_Completed_t2",
##   "Passes_Attempted_t2", "Touchdowns_t2", "Interceptions_t2"),
##   Ynodes = "Played_in_t3", abar = abar)
##
## Treatment Estimate:
##   Parameter Estimate: 0.25807
##   Estimated Std Err: 0.11371
##                   p-value: 0.023241
##   95% Conf Interval: (0.035194, 0.48095)
##
## Control Estimate:
##   Parameter Estimate: 0.16704
##   Estimated Std Err: 0.089568
##                   p-value: 0.06219
##   95% Conf Interval: (0, 0.34259)
##
## Additive Treatment Effect:
##   Parameter Estimate: 0.091031
##   Estimated Std Err: 0.14475
##                   p-value: 0.52943
##   95% Conf Interval: (-0.19268, 0.37474)
##
## Relative Risk:
##   Parameter Estimate: 1.545
##   Est Std Err log(RR): 0.69403
##                   p-value: 0.53081
##   95% Conf Interval: (0.39642, 6.0211)

```

```

##
## Odds Ratio:
##   Parameter Estimate: 1.7345
##   Est Std Err log(OR): 0.87585
##                   p-value: 0.52948
##   95% Conf Interval: (0.31164, 9.654)
##
## -      -      -
##
## OUTPUT: Maximal Model
## (abar = list(c(0, 1), c(0, 0)) )
##
## Estimator:  tmle
## Call:
## ltmle(data = data, Anodes = c(paste0(treats, "_t1"), paste0(treats,
##   "_t2")), Lnodes = c("Passes_Completed_t1", "Passes_Attempted_t1",
##   "Touchdowns_t1", "Interceptions_t1", "Passes_Completed_t2",
##   "Passes_Attempted_t2", "Touchdowns_t2", "Interceptions_t2"),
##   Ynodes = "Played_in_t3", abar = abar)
##
## Treatment Estimate:
##   Parameter Estimate: 0.35089
##   Estimated Std Err: 0.089785
##                   p-value: 9.3043e-05
##   95% Conf Interval: (0.17491, 0.52686)
##
## Control Estimate:
##   Parameter Estimate: 0.16704
##   Estimated Std Err: 0.089568
##                   p-value: 0.06219
##   95% Conf Interval: (0, 0.34259)
##
## Additive Treatment Effect:
##   Parameter Estimate: 0.18385
##   Estimated Std Err: 0.12635
##                   p-value: 0.14565
##   95% Conf Interval: (-0.063793, 0.43149)
##
## Relative Risk:
##   Parameter Estimate: 2.1006
##   Est Std Err log(RR): 0.59241
##                   p-value: 0.21024
##   95% Conf Interval: (0.65779, 6.7083)
##
## Odds Ratio:
##   Parameter Estimate: 2.6956
##   Est Std Err log(OR): 0.75234
##                   p-value: 0.18749
##   95% Conf Interval: (0.61696, 11.777)

```

Complete Statistics: Minimal Model

```
## OUTPUT: Minimal Model
## (abar = list(c(1, 1), c(0, 0)) )
##
## Estimator:  tmle
## Call:
## ltmle(data = data[, c("Years_Played_Before_t1", "Passes_Attempted_t1",
##      "Yards_Category_t1", "Passes_Attempted_t2", "Yards_Category_t2",
##      "Played_in_t3")], Anodes = c("Yards_Category_t1",
## "Yards_Category_t2"),
##      Lnodes = c("Passes_Attempted_t1", "Passes_Attempted_t2"),
##      Ynodes = "Played_in_t3", abar = abar)
##
## Treatment Estimate:
##   Parameter Estimate:  0.24926
##   Estimated Std Err:  0.044179
##   p-value:  1.6801e-08
##   95% Conf Interval: (0.16267, 0.33585)
##
## Control Estimate:
##   Parameter Estimate:  0.097432
##   Estimated Std Err:  0.072728
##   p-value:  0.18035
##   95% Conf Interval: (0, 0.23998)
##
## Additive Treatment Effect:
##   Parameter Estimate:  0.15183
##   Estimated Std Err:  0.085095
##   p-value:  0.074391
##   95% Conf Interval: (-0.014957, 0.31861)
##
## Relative Risk:
##   Parameter Estimate:  2.5583
##   Est Std Err log(RR):  0.7672
##   p-value:  0.22082
##   95% Conf Interval: (0.56872, 11.508)
##
## Odds Ratio:
##   Parameter Estimate:  3.0757
##   Est Std Err log(OR):  0.86007
##   p-value:  0.19145
##   95% Conf Interval: (0.56996, 16.597)
##
## - - -
## OUTPUT: Minimal Model
## (abar = list(c(1, 0), c(0, 0)) )
##
## Estimator:  tmle
## Call:
```



```

## ltmle(data = data[, c("Years_Played_Before_t1", "Passes_Attempted_t1",
##   "Yards_Category_t1", "Passes_Attempted_t2", "Yards_Category_t2",
##   "Played_in_t3")], Anodes = c("Yards_Category_t1",
##   "Yards_Category_t2"),
##   Lnodes = c("Passes_Attempted_t1", "Passes_Attempted_t2"),
##   Ynodes = "Played_in_t3", abar = abar)
##
## Treatment Estimate:
##   Parameter Estimate: 0.14206
##   Estimated Std Err: 0.080059
##   p-value: 0.075989
##   95% Conf Interval: (0, 0.29897)
##
## Control Estimate:
##   Parameter Estimate: 0.097432
##   Estimated Std Err: 0.072728
##   p-value: 0.18035
##   95% Conf Interval: (0, 0.23998)
##
## Additive Treatment Effect:
##   Parameter Estimate: 0.044628
##   Estimated Std Err: 0.10816
##   p-value: 0.6799
##   95% Conf Interval: (-0.16736, 0.25662)
##
## Relative Risk:
##   Parameter Estimate: 1.458
##   Est Std Err log(RR): 0.9353
##   p-value: 0.68682
##   95% Conf Interval: (0.23315, 9.1179)
##
## Odds Ratio:
##   Parameter Estimate: 1.5339
##   Est Std Err log(OR): 1.0562
##   p-value: 0.68544
##   95% Conf Interval: (0.19355, 12.156)
##
## - - -
## OUTPUT: Minimal Model
## (abar = list(c(0, 1), c(0, 0)) )
##
## Estimator:  tml
## Call:
## ltmle(data = data[, c("Years_Played_Before_t1", "Passes_Attempted_t1",
##   "Yards_Category_t1", "Passes_Attempted_t2", "Yards_Category_t2",
##   "Played_in_t3")], Anodes = c("Yards_Category_t1",
##   "Yards_Category_t2"),
##   Lnodes = c("Passes_Attempted_t1", "Passes_Attempted_t2"),
##   Ynodes = "Played_in_t3", abar = abar)
##

```

```
## Treatment Estimate:
##   Parameter Estimate: 0.44712
##   Estimated Std Err: 0.095931
##   p-value: 3.1495e-06
##   95% Conf Interval: (0.25909, 0.63514)
##
## Control Estimate:
##   Parameter Estimate: 0.097432
##   Estimated Std Err: 0.072728
##   p-value: 0.18035
##   95% Conf Interval: (0, 0.23998)
##
## Additive Treatment Effect:
##   Parameter Estimate: 0.34968
##   Estimated Std Err: 0.12021
##   p-value: 0.0036253
##   95% Conf Interval: (0.11409, 0.58528)
##
## Relative Risk:
##   Parameter Estimate: 4.589
##   Est Std Err log(RR): 0.77604
##   p-value: 0.049603
##   95% Conf Interval: (1.0027, 21.003)
##
## Odds Ratio:
##   Parameter Estimate: 7.4914
##   Est Std Err log(OR): 0.91247
##   p-value: 0.027319
##   95% Conf Interval: (1.2528, 44.798)
```